

NoodlesV: Learning from Corrections in Interactive Whole-Body PET/CT Segmentation

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Abstract. The AutoPET V baseline segments tumours from positron emission tomography and computed tomography (PET/CT) images with optional foreground and background scribbles. Its training scribbles are drawn from the reference tumour masks, whereas corrections during evaluation are placed on errors made by the model. This study generated new training examples from such errors and continued training with a loss adapted to the two PET tracers. On a held-out set, the resulting model almost doubled the baseline Dice score. On a separate 123-case evaluation, mean Dice increased from 0.650 before interaction to 0.745 after five corrections. A second model trained from random initialization showed the same response to corrections. The two-model ensemble gave the best available validation result and was selected for submission. Training on the type of correction expected at test time can improve a standard interactive segmentation model without changing its architecture.

Keywords: autoPET V · PET/CT · interactive segmentation · nnU-Net · corrective scribbles

1 Introduction

Whole-body PET/CT combines anatomical structure from computed tomography (CT) with tracer uptake from positron emission tomography (PET). Segmenting every tumour is difficult because lesions vary widely in size and appearance, and healthy organs can also show strong uptake. The problem changes with the tracer: ^{18}F -fluorodeoxyglucose (FDG) and prostate-specific membrane antigen (PSMA) have different normal uptake patterns.

Previous AutoPET entries established nnU-Net as a strong starting point and explored tracer-specific models, organ labels, anatomical pretraining and ensembles [6,9]. The AutoPET IV study by Huang et al. combined organ supervision with simulated user guidance and used different loss weights for FDG and PSMA [4]. Promptable PET/CT models have also compared alternative encodings of user input [8]. The present work follows this line rather than proposing a new network architecture.

The AutoPET V baseline receives four three-dimensional images: CT, PET, a foreground-scribble map marking missed tumour, and a background-scribble map marking tissue that should be removed from the prediction. During baseline

training, these scribbles are generated directly from the reference tumour masks [7]. During evaluation, however, each new scribble is placed on an error in the previous prediction. The baseline documentation identifies this difference as an open direction.

This study narrows that difference by training on corrections generated from model errors. It also uses tracer-specific loss weights and tests whether the response to correction is reproduced by a second model trained from scratch. The experiments ask three practical questions: whether the fine-tuned model improves on the organizer checkpoint, whether corrections improve successive predictions, and where the final system still fails.

2 Methods

2.1 Data and splits

The PSMA-FDG-PET-CT-Lesions v2 files were used for training [3]. The collection contains 1,611 labelled studies: 1,014 FDG and 597 PSMA. Local label inspection found no lesion in 513 FDG studies (50.6%).

The five-fold split used by the nnU-Net trainer was retained. Fold 0 contained 1,288 original training cases and 323 validation cases. Correction variants were added only when their parent scan was absent from the fold 0 validation set. The final continuation therefore trained on 1,452 entries: 1,288 original cases and 164 correction variants. A separate set of 123 cases was kept for the interactive evaluation. Fold 1 used its own 322-case validation set. No final model was trained on all 1,611 original cases.

2.2 Model and training

The organizer’s nnU-Net plan was kept unchanged [5]. It defines a six-stage three-dimensional `PlainConvUNet`, 32 to 320 feature maps, a $112 \times 160 \times 128$ -voxel patch and a target spacing of (3.0, 2.04, 2.04) mm. The four input channels are CT, PET and the two scribble maps. Foreground patches were sampled in 33% of training iterations.

Fold 0 started from the organizer checkpoint. The clean correction phase resumed from epoch 500 and ended at epoch 750. Fold 1 used the same architecture but started with random weights. It was first trained without model-error-derived corrections to epoch 1500, then continued with corrective examples. At epoch 1712, 171 new correction examples were added. Each contained a background scribble placed on a false-positive region; this changed the training examples, not the architecture or loss. Training ended at epoch 1850.

The loss combined the Dice loss, which measures overlap, with voxel-wise cross-entropy. For PSMA, Dice and cross-entropy were weighted 2:1; for FDG they were weighted 1:1. In this *tracer-specific loss weighting*, the balance between the two loss terms depends on the PET tracer. The choice follows the AutoPET IV design [4].

2.3 Training with corrective scribbles

An interaction begins with an automatic prediction made before the user provides a correction. This is round 0, and both scribble maps are empty. The prediction is compared with the reference mask. Missed tumour regions generate a foreground scribble; regions predicted as tumour outside the reference generate a background scribble. The official centreline, random and boundary strategies were used to draw these corrections. The points were stored directly in the two scribble channels, without Gaussian smoothing.

Correction examples for fold 0 were generated from scans belonging to the fold 0 training split. Before training, the parent identifier of every correction variant was compared with the validation identifiers. This check prevented a modified copy of a validation scan from entering training. The same rule was applied to fold 1.

2.4 Evaluation

Dice similarity coefficient (DSC) measures voxel overlap. The detection matching metric (DMM) is a lesion-level F1 score: predicted and reference connected components are paired when their intersection over union is at least 0.1. Both metrics were calculated with the challenge code.

Each case was evaluated before correction and after each of five simulated corrections. The area under each six-point curve (AUC) was calculated with the trapezoidal rule specified by the challenge [1]. Because the horizontal axis covers five intervals, the unnormalised AUC ranges from 0 to 5, not from 0 to 1. Dividing by five gives the mean score across the interaction trajectory. The challenge ranks AUC-DSC and AUC-DMM separately and gives the two ranks equal weight [1].

When a scan contains no reference tumour, DSC and DMM are undefined. The released reference loop does not add further simulated scribbles to these scans. A separate diagnostic analysis was therefore performed in which background scribbles were allowed to correct false detections. This analysis tests how the model responds to a clinician indicating a false positive; it is not used as an estimate of the official challenge score.

2.5 Compute and implementation

Initial experiments used an NVIDIA A100 with 80 GB of memory at approximately US\$1 per hour. Once the observed gains no longer justified that rate, training moved to an NVIDIA RTX A4500 with 20 GB of memory at approximately US\$0.25 per hour. Total cloud expenditure was approximately US\$220.

The final container averages the voxel logits from the fold 0 and fold 1 models, then returns the class with the larger average logit. A complete interaction takes about 32 seconds through the container code path. The image was tested locally without network access and within the 30 GB memory limit used for submission.

Table 1. Implementation summary requested by the AutoPET V method-description template.

Item	Choice	Details
Team and algorithm	Santiago Paramés-Estévez; NoodlesV	Independent submission
Pre-processing	nnU-Net v2	Resampling to (3.0, 2.04, 2.04) mm; CT and PET normalization; binary scribble maps
Post-processing	None	Binary class selected from the averaged logits
Augmentation	nnU-Net v2 defaults	Spatial and intensity transforms; 33% foreground oversampling
Test-time augmentation	None	Mirroring disabled
Ensemble	Two models	Fold 0 final checkpoint and fold 1 final checkpoint; voxel-logit average
Framework	nnU-Net v2.6.0; PyTorch 2.6	Six-stage 3D PlainConvUNet with four input channels
Loss	Dice plus cross-entropy	PSMA 2:1; FDG 1:1
Training data	1,014 FDG and 597 PSMA studies	Model-error-derived correction variants were restricted to training cases
Initialization	Mixed	Organizer checkpoint for fold 0; random initialization for fold 1
Hardware	RunPod A100 and A4500	80 GB and 20 GB video memory, respectively

The source code is available under the MIT License at <https://github.com/s-parames/NoodlesV>.

Table 1 follows the method-information template requested for AutoPET V submissions. The submitted algorithm is named *NoodlesV*.

3 Results

3.1 Automatic segmentation

On the 323 fold 0 validation cases, continued training increased mean DSC from 0.335 to 0.662 and pooled lesion-detection F1 from 0.566 to 0.651 (Table 2). The final model found 1,058 additional lesions, while producing 900 more false detections. The improvement therefore came mainly from greater sensitivity rather than greater specificity. Table 2 abbreviates matched detections as true positives (TP), false detections as false positives (FP), and missed lesions as false negatives (FN). On the separate 322-case fold 1 pool, the selected fold 1 checkpoint reached a round 0 DSC of 0.607 and DMM of 0.686. Averaging the final fold 0 and fold 1 logits increased these values to 0.629 and 0.702. This comparison informed model selection, but is not blind because fold 0 had seen cases from the fold 1 validation set during training.

Table 2. Automatic predictions before any correction. The first two rows use the same 323 fold 0 validation cases and include pooled lesion counts. The lower rows report the released evaluation metric on their stated pools; dashes mark counts that were not retained in the summary. The standalone fold 1 evaluation used its selected best checkpoint, whereas the ensemble used both final checkpoints.

Model	Pool	DSC	DMM	TP	FP	FN
Organizer baseline	Fold 0, 323	0.335	0.566	2,815	1,590	2,723
Final fold 0 model	Fold 0, 323	0.662	0.651	3,873	2,490	1,665
Selected fold 1 model	Fold 1, 322	0.607	0.686	–	–	–
Final ensemble	Fold 1, 322	0.629	0.702	–	–	–

Table 3. Interactive results for both single models and the submitted ensemble. Fold 0 and the ensemble use the same 75 lesion-positive cases from the 123-case pool. Fold 1 uses 218 lesion-positive cases from its separate 322-case validation set. Stronger cell shading indicates a higher value on one common scale; colour identifies the model.

Model	Metric	0	1	2	3	4	5
Fold 0	DSC	0.650	0.706	0.719	0.728	0.739	0.745
	DMM	0.610	0.657	0.659	0.667	0.681	0.693
Fold 1	DSC	0.607	0.629	0.638	0.648	0.653	0.655
	DMM	0.589	0.613	0.614	0.625	0.626	0.628
Ensemble	DSC	0.670	0.726	0.739	0.748	0.757	0.756
	DMM	0.624	0.667	0.675	0.694	0.689	0.675

3.2 Response to correction

The 123-case interactive set contained 75 scans with lesions and 48 without lesions. For the lesion-positive scans, fold 0 mean DSC increased from 0.650 before correction to 0.745 after five corrections (Table 3). More than half of the total improvement followed the first scribble. Mean per-case DMM increased from 0.610 to 0.693. The unnormalised AUC-DSC was 3.589 (0.718 after division by five), while AUC-DMM was 3.315 (0.663 after division by five).

Figure 1 compares the two single models and the ensemble. Panel (a) measures overlap when lesions are present. Panel (b) gives the percentage of lesion-free scans for which no tumour is predicted; higher values indicate better case-level specificity. Panel (c) counts the remaining disconnected false-positive regions. Fold 1 has fewer such regions, but its curve comes from a different validation set and should not be read as a direct advantage over the other two models.

The ensemble had the highest DSC at every round on the shared 123-case pool. Its AUC-DSC was 3.683 (0.737 after division by five) and its AUC-DMM was 3.374, compared with 3.589 (0.718) and 3.315 for fold 0. It also cleared more lesion-free scans. These results motivated its selection for submission. The

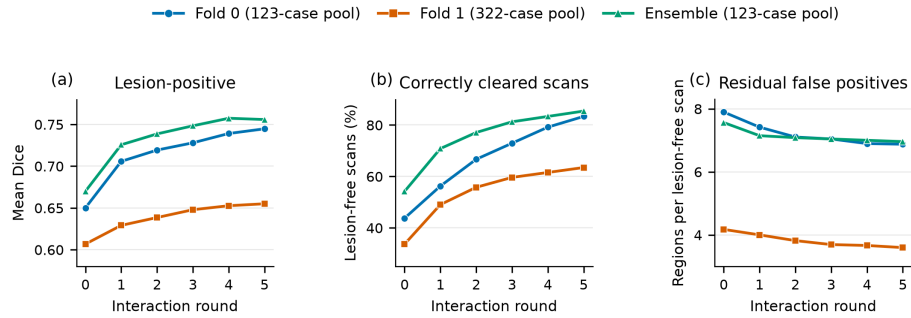


Fig. 1. Response over six interaction rounds. (a) Mean DSC measures overlap in lesion-positive scans. (b) Correctly cleared scans measure case-level specificity. (c) The mean number of false-positive components shows how fragmented the remaining errors are. Fold 0 and the ensemble use the same 123-case pool; fold 1 uses its separate 322-case validation set.

comparison is useful for model selection, but it is not a blind estimate: one component had seen cases that occur in the other fold’s validation set.

Figure 2 shows four lesion-positive examples from the preserved fold 0 case figures. The axial and sagittal slices include both tracers and lesions of different extent. They illustrate where prediction and reference agree, but were not used for quantitative model selection.

3.3 Training history and second-fold evaluation

Figure 3 shows the complete logged history. Fold 0 began from the organizer checkpoint and ended at epoch 750. Fold 1 began from random weights and first trained without model-error-derived corrections. Corrective training ran from epoch 1500 to 1850. At epoch 1712, 171 additional examples were introduced, each forcing a background correction on a false-positive region. This was an attempt to teach the model how to remove excess predictions. The architecture, loss and optimizer were unchanged.

The fold 1 best checkpoint was evaluated on 218 lesion-positive cases from its 322-case validation set. DSC increased from 0.607 to 0.655, and DMM from 0.589 to 0.628 (Table 3). This reproduced the direction of the fold 0 result after training from random initialization. The ensemble instead uses the final checkpoint from each fold.

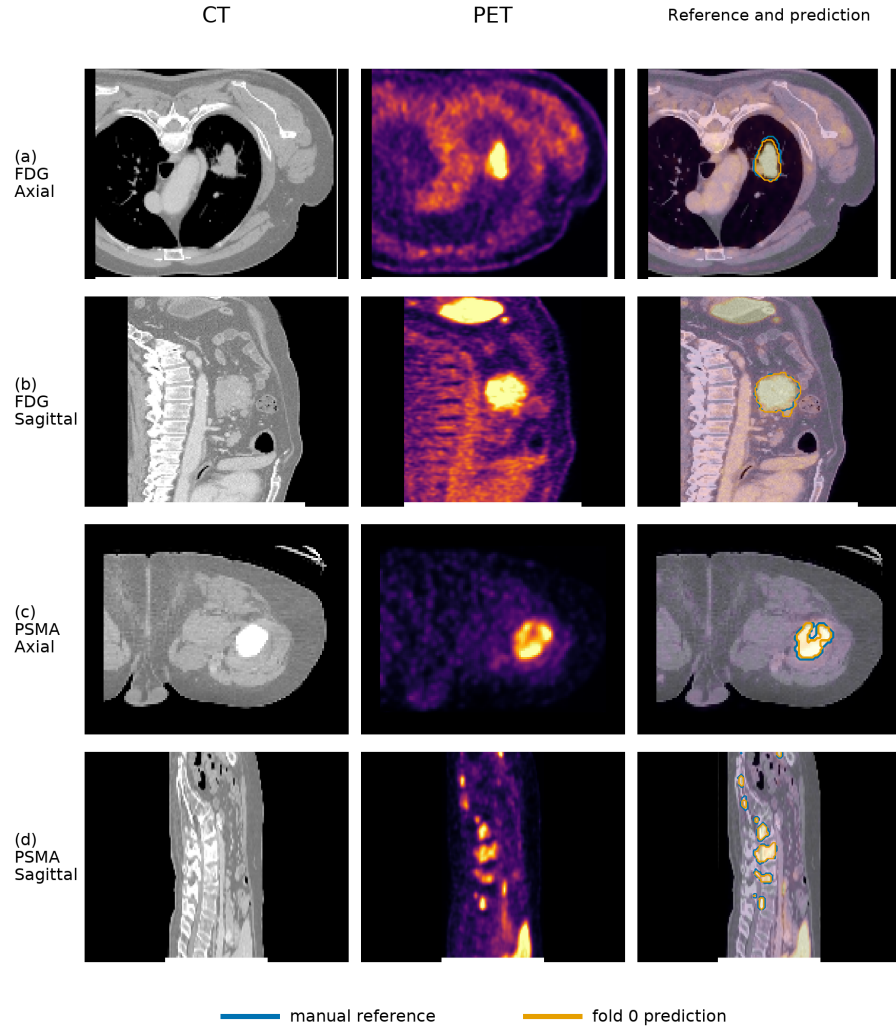


Fig. 2. Four lesion-positive cases shown in two orientations: (a) axial FDG, (b) sagittal FDG, (c) axial PSMA and (d) sagittal PSMA. Columns show CT, PET, and the overlay of the manual reference (blue) with the final fold 0 prediction (orange). These examples are illustrative rather than a representative sample; performance is quantified over the complete evaluation pools in Table 3.

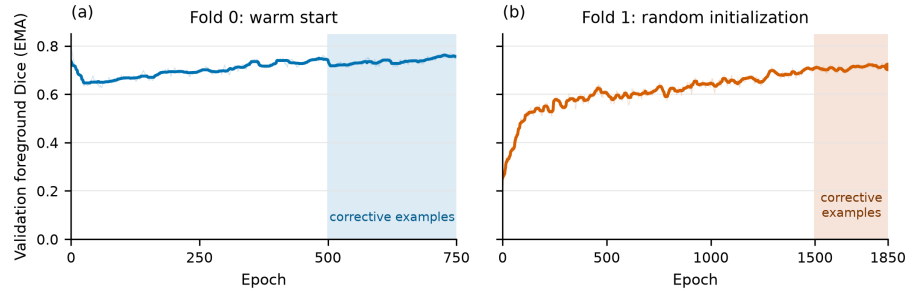


Fig. 3. Training histories for (a) fold 0 and (b) fold 1. Light lines show the logged validation foreground Dice exponential moving average (EMA), and dark lines show its 25-epoch rolling median. Shaded regions mark training with model-error-derived corrections; fold 1 training ends at epoch 1850. The background-correction examples introduced at epoch 1712 are described in the text.

4 Discussion

Training on model errors improved both the automatic prediction and the response to later corrections. The largest interactive gain followed the first scribble, which is useful when a clinician has little time to edit a scan. The result appeared in both folds despite their different initialization. These are internal validation results, not a claim of a new state of the art. No AutoPET V test score was available at the time of writing.

The scale of a Dice score depends strongly on the cohort, lesion burden and annotation protocol. For context, Kalisch et al. reported cross-validation Dice values of 0.769 for FDG and 0.613 for PSMA in AutoPET III [6]. A lymphoma study reported a median manual inter-observer Dice of 0.84 [2]. Neither result is directly comparable with the mixed-tracer AutoPET V interaction curves, but both show why a single Dice value should not be described as human-level or clinically sufficient.

False positives are the clearest remaining problem. Correction removed all predicted lesions from many lesion-free scans, but a small subset retained several disconnected regions. This agrees with the known difficulty of separating tumour uptake from normal organ uptake. The increased number of false detections in Table 2 also shows that the gain in sensitivity was not free.

Anatomical supervision was explored because it had helped previous AutoPET systems [6,4]. TotalSegmentator [10] produced masks for 14 selected organs in 1,591 studies. An additional output of the network was trained to reproduce those masks, with the intention that shared image features would encode organ location. In two runs where this organ loss updated the lesion network, the model ended up labelling every voxel as non-lesion. The final runs therefore blocked the organ loss from changing the lesion network and removed the organ

output at inference. Organ information cannot have improved the submitted predictions in this configuration.

This negative result does not contradict the earlier papers. Kalisch et al. used tracer-specific ensembles, 117 anatomical labels or carefully selected tracer-specific subsets, and explicit weighting of lesion and anatomy classes. Their results also show that using all organ labels without appropriate weighting could reduce Dice [6]. The present experiment used only 14 organs, a separate auxiliary output and a different optimization schedule. The most likely explanation is therefore implementation and loss balance, not that anatomical context is useless. No ablation was completed to identify one cause.

Other limitations remain. Corrective training and tracer-specific loss weighting changed together, so their individual contributions cannot be separated. Simulated scribbles may not match the way clinicians mark errors. The ensemble comparisons reuse cases seen by one of the two component models. Finally, the additional lesion-free experiment is a response test, not part of the official simulated evaluation.

The container test addressed deployment only. It confirmed that the packaged ensemble reads the expected inputs, writes a geometrically valid binary mask, runs without network access and stays within the memory limit. Accuracy on the hidden test set remains to be measured by the challenge.

5 Conclusion

Training with tracer-specific loss weights and corrections drawn from model errors substantially improved the organizer checkpoint on the held-out fold. A second model reproduced the response to interaction, and the two-model ensemble gave the strongest available validation result. False positives in lesion-free scans remain the central weakness. An interactive model benefits when its training examples resemble the corrections it will receive.

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